Ajay Singh

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Professional Summary

Artificial Intelligence and Data Science undergraduate with hands-on experience building AI/ML applications in Python, including Retrieval-Augmented Generation (RAG), LLM-powered agents, resume analysis, semantic search, vector databases, and secure tool-calling workflows. Practical AI engineering with a focus on building reliable and production-oriented applications.

Education

Ajeenkya DY Patil University Pune, India
Bachelor of Engineering (B.E.) in Artificial Intelligence and Data Science *2023 – 2027*
- Class XII: 93%

Projects

Candidate Analysis Agent | **Python, LangChain, Google Gemini, Qdrant, Hugging Face, RAG**
- Built a Retrieval-Augmented Generation (RAG) system that processes resume PDFs, splits documents into chunks, generates semantic embeddings, and stores them in a Qdrant vector database.
- Implemented semantic similarity retrieval to identify relevant resume sections and used Google Gemini to generate context-grounded answers from retrieved information.

Secure Customer Support AI Agent | **Python, LangChain, MCP, Groq, Descope**
- Developed an AI customer support agent using LangChain and Model Context Protocol (MCP) to securely access customer, order, product, and refund tools.
- Integrated Descope authentication and scope-based authorization to control access to protected tools while using verified tool results for customer and refund operations.

AI Resume Analyzer | **Python, Google Gemini, Pydantic, PDF Processing**
- Built an AI resume analysis pipeline that extracts resume text from PDF files and compares candidate resumes against job descriptions using Google Gemini.
- Implemented structured JSON validation with Pydantic to generate a consistent match score, identify strong skills, highlight missing skills, and recommend resume improvements.

SkillSync – Career Tracking Backend Application | **FastAPI, SQLite, SQLAlchemy** Mar 2026 – Present
- Developed a FastAPI backend application for tracking user skills and career growth through RESTful APIs for adding and retrieving skill information.
- Integrated SQLite with SQLAlchemy for persistent data storage and structured the application for scalable backend development, with ongoing work on authentication and intelligent recommendation features.

Technical Skills

Languages: Python, SQL

AI/ML: Machine Learning, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Semantic Search, Prompt Engineering

AI Tools: LangChain, Google Gemini, Hugging Face Sentence Transformers, Qdrant, Model Context Protocol (MCP), Groq, Descope

Backend/Databases: FastAPI, REST APIs, SQLite, SQLAlchemy, MySQL

Cloud/DevOps: Docker, AWS CloudFormation

Core Concepts: Data Preprocessing, Exploratory Data Analysis (EDA), Model Training, Model Evaluation, Vector Databases, Information Retrieval, Authentication, Authorization

Training / Certifications

Machine Learning Basics Using Scikit-learn (Self-learning) Jan 2026 – Feb 2026
Online / Virtual
- Studied supervised learning, regression, classification, K-Nearest Neighbors (KNN), Logistic Regression, Linear Regression, Random Forest, model training, and evaluation metrics.

Python for Data Science (Self-learning) Jan 2026 – Feb 2026
Online / Virtual
- Practiced Python with NumPy, Pandas, and Matplotlib, including data cleaning, data preprocessing, and exploratory data analysis (EDA).